

Workshop on Evaluation of Protective Textile

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1. INTRODUCTION

The advancement of textile technology has led to the development of sophisticated machinery for manufacturing and testing high-performance fabrics. This report presents an overview of key equipment used in modern textile laboratories and industries, including their purpose, technical specifications, testing standards, and applications.

2. ELECTRONIC JACQUARD MACHINE

2.1 Purpose

The Electronic Jacquard Machine is a fully automated weaving system designed to create complex and precise woven patterns through electronic control of individual warp yarns.

2.2 Figure 1: Electronic Jacquard Machine



2.3 Technical Parameters

- Shuttleless Rapier Loom of 170 cm reed width with colour selectors
- Fully electronic Jacquard with 900 hooks

2.4 Testing / Quality Considerations

- Dimensional stability
- Colorfastness
- Strength and uniformity

2.5 Applications & Research Areas

- Applications: Apparel, upholstery, smart textiles, composites
- Research Areas: Advanced weaving, e-textiles

2.6 Vendor

Paramount Machineries

3. COATING & LAMINATION MACHINE (WITH ADHESION TESTING)

3.1 Purpose

Used to apply coatings or laminated layers to textiles to enhance waterproofing, durability, and protection.

3.2 Figure 2: Coating & Lamination Machine



3.3 Technical Parameters

- Size: 450 mm × 450 mm
- Knife-on-Roll & Knife-on-Air system with dryer
- Lamination attachment with automatic control

3.4 Testing Standards

- ISO 2411
- ISO 811 / AATCC 127

3.5 Applications & Research Areas

- Applications: PPE, automotive, medical textiles
- Research Areas: Functional coatings, multilayer textiles

3.6 Vendor

Tuskar Engineering

4. HYDROSTATIC PRESSURE TESTER

4.1 Purpose

Measures resistance of fabric to water penetration under increasing pressure.

4.2 Figure 3: Hydrostatic Pressure Tester



4.3 Technical Parameters

- Pressure range: 0–1000 kPa
- Pressure rate: 60 kPa/min
- Test head: 100 cm²
- Automatic clamping

4.4 Testing Standards

- ISO 811
- AATCC TM 127

4.5 Applications & Research Areas

- Applications: Rainwear, tents, medical textiles
- Research Areas: Waterproof fabrics

4.6 Vendor

Vital Technologies

5. CUT RESISTANCE TESTER

5.1 Purpose

Evaluates resistance of textile materials to cutting forces.

5.2 Figure 4: Cut Resistance Tester



5.3 Technical Parameters

- Load: ± 0.005 N
- Stroke: 50 mm
- Speed: 10 cm/sec

5.4 Testing Standards

- ISO 13997
- ASTM F2992

5.5 Applications & Research Areas

- Applications: Protective gloves, industrial wear
- Research Areas: High-performance fibers

5.6 Vendor

Vital Technologies

6. CONCLUSION



These machines play a crucial role in textile manufacturing and testing by ensuring quality, performance, and innovation across various applications.
